An Analysis of the All or None Model

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We analyze the all or none model, where new nodes attach to a randomly selected node, and with probability p connect to all of the selected node's neighbors. We find a critical point when $p = 0.5$, where the structure of the graph changes. We derive analytical expressions for the expected number of edges, degree distribution, clustering, assortativity, mean shortest path length, and clique distribution. These predictions are validated through numerical simulations.

I. INTRODUCTION

Networks are widely used to model many systems in physics, biology, and social science. Using simple growth rules like the all or none model to study properties of large limit networks can be extremely useful. Clustering models, like the all or none model, can be especially important as most real world networks aren't random, and have strong local structure. The all or none model can be viewed as a simplified version of some of the more widely studied copying based models that give good clustering.

The all or none model is an undirected graph that starts a single node. At each time step, a random node is chosen and the new node is attached. Then, with probability p, which is chosen at the beginning, the new node is also connected to all of the selected node's neighbors. This simplistic model produces different structural behaviors depending on the chosen p which are studied in this paper.

In this paper, we develop analytical models of the total number of edges, degree distribution, clustering, assortativity, mean shortest path, and clique distribution in the all or none model. We back up each of these analytical approximations with numerical data showing that the chosen analyses are accurate. We also show that $p = 0.5$ is the critical point on which the structure of the resulting graph drastically changes.

II. ANALYTICAL DERIVATIONS

A. Total Number of Edges

This recursion is the simplest recursion. We have zero edges with one node, and one edge with two. When we add a new node, we always add one edge. We then also add with probability p add whatever the degree of the node that we select. The expected degree of a randomly selected node at n is

$$\frac{2N(n)}{n} \tag{1}$$

Therefore the master equation for the total number of edges in the all or none model is

$$N(n+1) - N(n) = 1 + \frac{2pN(n)}{n} \tag{2}$$

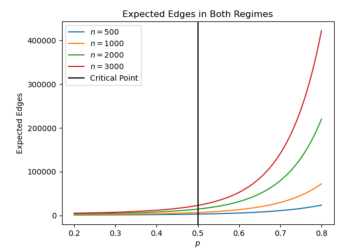

FIG. 1. Expected Edges for Different p, n

We plot this for multiple n and p in FIG. 1. As we will derive in the next section, the critical point of $p = 0.5$ can be seen quite clearly in this graph, as the number of edges exponentially takes off after that critical point.

B. Degree Distribution

We first must write down the master equation $N_k(n)$ for the degree distribution. This will obviously have a base case of $N_k(1) = \delta_{1,k}$This is quite easy, as we increase the amount of expected nodes of degree k by one if we connect to a node of degree $k-1$, or by two if we connect and copy. We must remove a node if we connect to a node of degree k. This leads to the following simple recurrence relation.

$$D_k(n+1) - D_k(n) = \delta_{1,k} + \frac{1+p(k-1)}{n} D_{k-1}(n) \tag{3}$$

$$-\frac{1+pk}{n} D_k(n) \tag{4}$$

We can then convert this to a degree probability distribution P_k to get

$$(2+pk)P_k = \delta_{1,k} + [1+p(k-1)]P_{k-1} \tag{5}$$

It is clear that $P_1 = \frac{1}{2+p}$, so we can rewrite P_k as

$$P_k = \begin{cases} \frac{1}{2+p}, \, k = 1 \\ \frac{1+p(k-1)}{2+pk}, \, else \end{cases} \tag{6}$$

$$= P_1 \prod_{j=2}^{k} \frac{1+p(j-1)}{2+pj} \tag{7}$$

$$= P_1 \prod_{j=2}^{k} \frac{j-1+\frac{1}{p}}{j+\frac{2}{p}} \tag{8}$$

We can then examine the numerator and denominator separately, and replace them with gamma functions.

$$\prod_{j=2}^{k} j-1+\frac{1}{p} = \prod_{m=1}^{k-1} m+\frac{1}{p}, \, m = j-1 \tag{9}$$

$$= \frac{\Gamma(k+\frac{1}{p})}{\Gamma(1+\frac{1}{p})} \tag{10}$$

$$\prod_{j=2}^{k} j+\frac{2}{p} = \frac{1}{1+\frac{2}{p}} \prod_{j=1}^{k} j+\frac{2}{p} = \frac{\Gamma(k+1+\frac{2}{p})}{(1+\frac{2}{p})\Gamma(1+\frac{2}{p}} \tag{11}$$

$$= \frac{\Gamma(k+1+\frac{2}{p})}{\Gamma(2+\frac{2}{p})} \tag{12}$$

Now, we can rewrite the degree probability distribution, and use the property of the ratios of gamma functions to derive a power law tail for the degree distribution, which will give us the critical exponents.

$$P_k = \frac{1}{2+p} \frac{\Gamma(k+\frac{1}{p})}{\Gamma(1+\frac{1}{p})} \frac{\Gamma(2+\frac{2}{p})}{\Gamma(k+1+\frac{2}{p})} \tag{13}$$

$$= \frac{1}{2+p} \frac{\Gamma(k+\frac{1}{p})}{\Gamma(k+1+\frac{2}{p})} \frac{\Gamma(2+\frac{2}{p})}{\Gamma(1+\frac{1}{p})} \tag{14}$$

$$\approx k^{\frac{1}{p}-1-\frac{2}{p}} \tag{15}$$

$$\approx k^{-(1+\frac{1}{p})} \tag{16}$$

Since the critical exponent, we see that $p = \frac{1}{2}$ is the critical exponent, and $p < \frac{1}{2}$ is the well defined regime while $p > \frac{1}{2}$ has a fat tail.

C. Triangles and Clustering

The first time we can have a triangle is when $n = 3$, so our base case is zero triangles when $n < 3$. In the no-copy case, we can never add a triangle, so we only have to worry about what happens in the copy case. There are two contributions to the added number of triangles, the degree of the chosen node, and the triangles of the chosen node. This gives the master equation for the total number of expected triangles as

$$T(n+1) - T(n) = p\left(\frac{2N(n)}{n} + \frac{3T(n)}{n}\right) \tag{17}$$

Now, all we need for the clustering is to find the number of trails of length three in the graph. When we add a node, we need to increase the number of trails by the number of trails that this node is expected to be a part of. In the no-copy case, this is simply the degree of the chosen node. In the copy case, there is a contribution from when the new node attaches somewhere in a trail of three, which gets a factor of three since it could attach anywhere in that trail to "extend" it. Finally, in the copy case we have a combinatorial contribution formed from the edges around the selected node to get the following equation

$$W(n+1) - W(n) = \frac{2N(n)}{n} + \frac{p}{n}\left[3W(n) + 4N(n)\right] \tag{18}$$

Finally, to get the clustering we simple multiply the number of triangles by three in order to account that each node in a triangle counts as a separate triangle to get

$$C(n) = \frac{3T(n)}{W(n)} \tag{19}$$

D. Degree Assortativity

In order to analytically get the assortativity of the all or none model, we first construct the stub matrix $E(n)$, where $E_{ij}(n)$ is the number of edges between nodes of degree i and j at time n. However, the diagonal of the stub matrix is doubled, as it is convenient. Through the stub matrix, we are able to easily calculate the assortativity, and we can define a closed recurrence of the stub matrix. We can get the degree distribution from the stub matrix by

$$d_k(n) = \frac{1}{k} \sum_j E_{kj}(n) \tag{20}$$

We are looking to obtain $E_{ij}(n+1) - E_{ij}(n)$, as this would allow us to obtain the stub matrix at any time since the base case is quite simple, where at $n = 2$, $E_{ij} = 2\delta_{i,1}\delta j, 1$. In order to do this, we notice that when we add a new node into the graph, we select a node of degree k with probability $\frac{d_k}{n}$. The degree of the selected node always increases by one, and in turn, the stubs connected to the selected node also increase. With probability p, the new node also connects to all of the nodes' neighbors, which increases the degree of all the stubs connected to the neighbors of the selected nodes. For updating the selected node and adding the new node, it is easiest to deal

with specific elements of the stub matrix. We can divide this into three major cases at each iteration: reclassifying the selected node ($E^{(r)}$), the no-copy case ($E^{(nc)}$), and the copy case ($E^{(cc)}$). The master equation is then

$$E_{ij}(n+1)-E_{ij}(n) = E_{ij}^{(r)}(n)+E_{ij}^{(nc)}(n)+E_{ij}^{(cc)}(n) \quad (21)$$

We always reclassify the selected node, as we always add a single edge to it whether or not we copy its other edges. This gives us the following change in

$$E_{ij}^{(r)}(n) = \frac{E_{i,j-1} + E_{i-1,j} - 2E_{ij}}{n} \quad (22)$$

For the no-copy case, for a selected node of degree i, all we need to do is increase number of nodes of degree i to nodes of degree one by the proportional number of nodes of degree $i-1$.

$$E_{ij}^{(nc)}(n) = \frac{(1-p)}{n}(d_{i-1}\delta_{1,j} + d_{j-1}\delta_{1,i}) \quad (23)$$

The copying case is where the real complexity comes in. The edges from nodes of degree i to degree j first increase proportional to the number of edges between nodes of degree $i-1$ to degree $j-1$, as the selected node increases, as well as the neighbor. We also need to account for all of the neighbors that get increased, which is why we need to iterate through each neighbor as well. This gives the contributing term from the copy case as

$$\begin{aligned} E_{ij}^{(cc)} = \frac{p}{n} \times \Bigg[& E_{i-1,j-1}(n) + \\ & E_{ij}(n) \sum_k \frac{E_{ik}(n)}{d_i} \left[E_{i-1,k}(n) - E_{i,k}(n)\right] + \\ & E_{ij}(n) \sum_k \frac{E_{jk}(n)}{d_j} \left[E_{j-1,k}(n) - E_{j,k}(n)\right] \Bigg] \end{aligned} \quad (24)$$

To calculate the assortativity, we have a formula that only needs the stub matrix we calculated. In order to adapt it to the stub matrix we do the following steps.

$$r = \frac{\langle k_i k_j \rangle - \langle k \rangle^2}{\langle k^2 \rangle - \langle k \rangle^2} \quad (25)$$

$$r = \frac{\sum_{i,j} \frac{ijE_{ij}}{m} - \left(\sum_{i,j} iE_{ij}\right)^2}{\sum_{i,j} i^2 E_{ij} - \left(\sum_{i,j} iE_{ij}\right)^2} \quad (26)$$

E. Average Shortest Path

Let

$$D(n) = \sum_{i<j} d_n(i,j) \quad (27)$$

be the total pair-wise distance of the network at time n where $d_n(i,j)$ is the individual shortest path between nodes i, j. The average shortest path is then

$$L(n) = \frac{2D(n)}{n(n-1)} \quad (28)$$

Now, the path from the new node to the selected node is always one. In the no-copy case, the distance from the new node to every other node is the same as the distance from the selected node to every other node plus one. In the copy case, we do not add one to the distance from the selected node to every other node. Therefore, our recurrence for the total distance in the graph is

$$D(n+1) - D(n) = \frac{2D(n)}{n} + (1-p)n + p \quad (29)$$

$$= \frac{2D(n)}{n} + n - p(n-1) \quad (30)$$

We solve this recurrence by setting $a_n = \frac{D(n)}{n(n+1)}$ and using the base case of $a_2 = \frac{1}{6}$ to get

$$a_{n+1} = a_n + \frac{n - p(n-1)}{(n+1)(n+2)} \quad (31)$$

$$a_n = a_2 + \sum_{k=2}^{n-1} \left(\frac{2p-1}{k+1} + \frac{2-3p}{k+2} \right) \quad (32)$$

We can further simplify this using the harmonic numbers to get

$$a_n = (1-p)H_n + \frac{5p-4}{2} + \frac{2-3p}{n+1} \quad (33)$$

$$D(n) = n(n+1)\left[(1-p)H_n + \frac{5p-4}{2} + \frac{2-3p}{n+1}\right] \quad (34)$$

$$L(n) = \frac{2n(n+1)}{n-1}\left[(1-p)H_n + \frac{5p-4}{2} + \frac{2-3p}{n+1}\right] \quad (35)$$

F. Clique Distribution

In addition to the earlier analyses, we also choose to analyze the distribution of the cliques. A clique is a subset of the network such that every node is connected. For the master equation, we split the recurrence into cliques of size one, two, and greater than two. Since each node is itself a clique of size one, the number of expected cliques of size one always increases by just one.

A clique of size two always appears when a new node is added with the new node and the selected node. In the copy case, the new node also forms a clique of size two with the selected node's neighbors, so it increases by one plus the average degree of the network. For the recurrence for cliques greater than two, we can add a clique of size k if the selected node is in a clique of size k or $k-1$. The master equation is therefore

$$K_1(n+1) - K_1(n) = 1 \tag{36}$$

$$K_2(n+1) - K_2(n) = 1 + \frac{2p}{n} K_2(n) \tag{37}$$

$$K_{i>2}(n+1) - K_{i>2}(n) = \frac{p}{n}\left[iK_i(n) + (i-1)K_{i-1}(n)\right] \tag{38}$$

III. NUMERICAL SIMULATIONS

A. Creation

The all or none model is easy to generate from the adjacency matrix. First, we generate an empty $n \times n$ matrix for any chosen n. The diagonal will always be zero in this model. We start at $n = 2$, and randomly select another integer k from $1..n$, not including n. We choose a random number in the range of $(0, 1)$, and if it is less than p we copy both the row i and column i to the row and column of n respectively. We then set create an undirected edge from i to n in the adjacency and repeat until the entire adjacency matrix is completed.

For all of the numerical results, we can simply run the creation algorithm a certain number of tries and average the outputs of the specific metrics. We use NetworkX[1] to quickly and accurately calculate all of the metrics studied in this analysis. All of the code used in order to calculate the numerical results is provided in the Jupyter Notebook.

B. Degree Distribution

We can use the degree function to create a histogram of the degrees in a given network in NetworkX. In FIG. 2, we show the analytical result in orange and the numerical result in blue. We see that the analysis is extremely accurate in both regimes and leads to wildly different structures.

C. Triangles and Clustering

We can find the number of triangles for a given graph by using NetworkX. In FIG. 3, we compare the expected versus the numerical number of triangles expected for $n = 500$ at a multitude of different p.

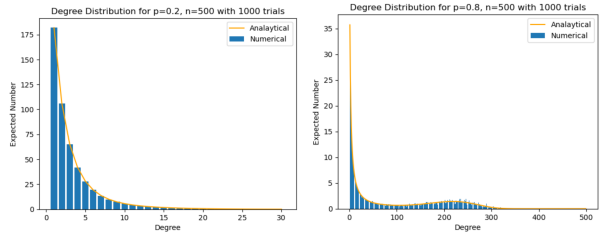

FIG. 2. Degree Distribution

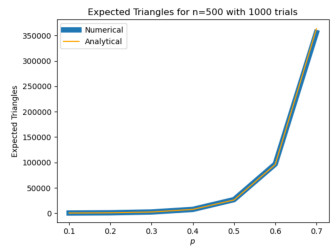

FIG. 3. Expected Triangles

The global clustering can be found in NetworkX using the transitivity, and we do a similar comparison to the number of triangles where we compare for multiple p for $n = 500$. Now, we can see in FIG. 4 that the analytical approximation overshoots the numerical results. This is because we approximate the global clustering as proportional to the expectation of the triangles over the expectations of the trails, but the actual value is the expectation of the triangles over the trails, so this approximation works quite well at low p but gets progressively worse at larger values.

$$C(n) = \frac{3T(n)}{W(n)} = \frac{3\mathbb{E}\left[T(n)\right]}{\mathbb{E}\left[W(n)\right]} \tag{39}$$

$$\neq \mathbb{E}\left[\frac{3T(n)}{W(n)}\right] \tag{40}$$

D. Assortativity

Since we calculated the expected assortativity through the expected stub matrix, the approximation isn't ex-

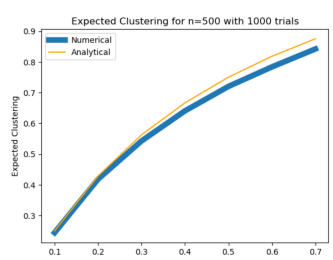

FIG. 4. Global Clustering

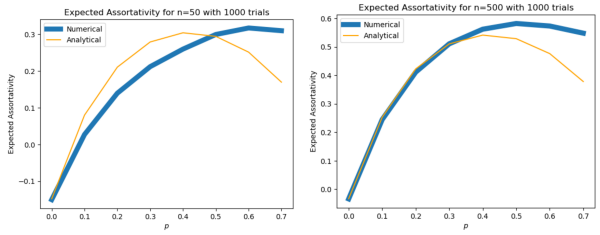

FIG. 5. Assortativity

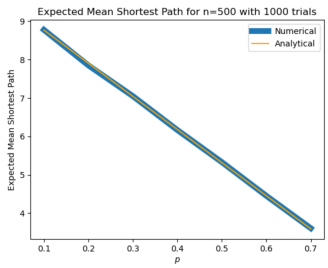

FIG. 6. Mean Shortest Path

act. This is because calculating the assortativity from the stub matrix is quadratic, so this approximation isn't as good when n is small. Unfortunately, calculating assortativity when n is large is takes quite a long time, so we show that the accuracy of our analytical result improves from when $n = 50$ to $n = 500$ in FIG. 5. We can see that the approximation is quite good in the sparse regime, but drops off significantly after the critical point due to the different structure with higher clustering.

E. Average Shortest Path

Again, numerically we can average the value of a bunch of different graphs using NetworkX. Comparing against the analytical solution for multiple different p when $n = 500$ we get FIG. 6.

F. Clique Distribution

In NetworkX, to get the distribution of the cliques, we can enumerate all cliques and manually count the number of cliques of different sizes. This, unfortunately, takes a super long time so we can only compare with a very small network of $n = 20$. Our analytical recurrence provides a good model of the network in both regimes as we can see in FIG. 7.

IV. CONCLUSION

In this paper, we analyze the all or none model, and derive analytical expressions for many of structural properties including degree distribution, clustering, assortativity, mean shortest path, and clique distribution. We show that the analytical expressions correctly predict the behavior of the numerical simulations of the all or none model.

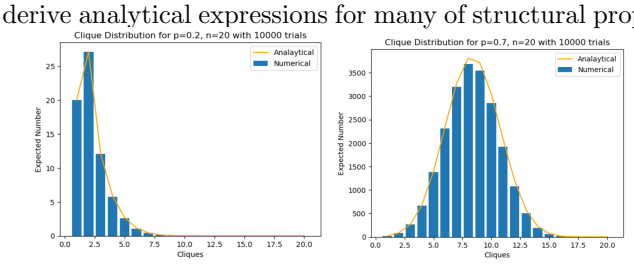

FIG. 7. Clique Distribution

The key result of this paper is deriving the critical point of $p = 0.5$, where the structural properties of the networks below and above this critical point is very different. When $p < 0.5$, the network remains relatively sparse, while when $p > 0.5$, the graph becomes extremely dense. Depending on the choice of p, we can wildly vary the behavior of the networks generated by the all or none model.

The all or none model provides a great simple model for studying the structure and clustering of networks in copying models. Future work could be to derive analytical expressions for the local clustering, or extend this model with different variations with partial copying.

[1] A. A. Hagberg, D. A. Schult, and P. J. Swart, Exploring network structure, dynamics, and function using networkx, Python in Science Conference 10.25080/TCWV9851 (2008).